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Cell Calcium. Author manuscript; available in PMC 2023 March 01.

Published in final edited form as:

Cell Calcium. 2022 March ; 102: 102538. doi:10.1016/j.ceca.2022.102538.

Channelopathy-causing mutations in the $S_{45}A/S_{45}B$ and HA/HB helices of $K_{Ca}2.3$ and $K_{Ca}3.1$ channels alter their apparent Ca^{2+} sensitivity

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Abstract

Small- and intermediate-conductance Ca^{2+}-activated potassium ($K_{Ca}2.x$ and $K_{Ca}3.1$) channels are activated exclusively by a Ca^{2+}-calmodulin gating mechanism. Wild-type $K_{Ca}2.3$ channels have a Ca^{2+} EC_{50} value of ~0.3 μM, while the apparent Ca^{2+} sensitivity of wild-type $K_{Ca}3.1$ channels is ~0.27 μM. Heterozygous genetic mutations of $K_{Ca}2.3$ channels have been associated with Zimmermann-Laband syndrome and idiopathic noncirrhotic portal hypertension, while $K_{Ca}3.1$ channel mutations were reported in hereditary xerocytosis patients. $K_{Ca}2.3_S436C$ and $K_{Ca}2.3_V450L$ channels with mutations in the $S_{45}A/S_{45}B$ helices exhibited hypersensitivity to Ca^{2+}. The corresponding mutations in $K_{Ca}3.1$ channels also elevated the apparent Ca^{2+} sensitivity.

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Author contributions

H.M.N., H.W. and M.Z. conceptualized the project. R.O., M.A.R., and M.Z. undertook electrophysiology studies. Y.W.N., H.M.N and G.Y. undertook molecular biology and cell culture studies. M.C. performed computational work. All authors contributed to the manuscript and the figures.

Razan Orfali and Young-Woo Nam contributed equally to this work.

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Conflict of interest

none.

CRediT author statement

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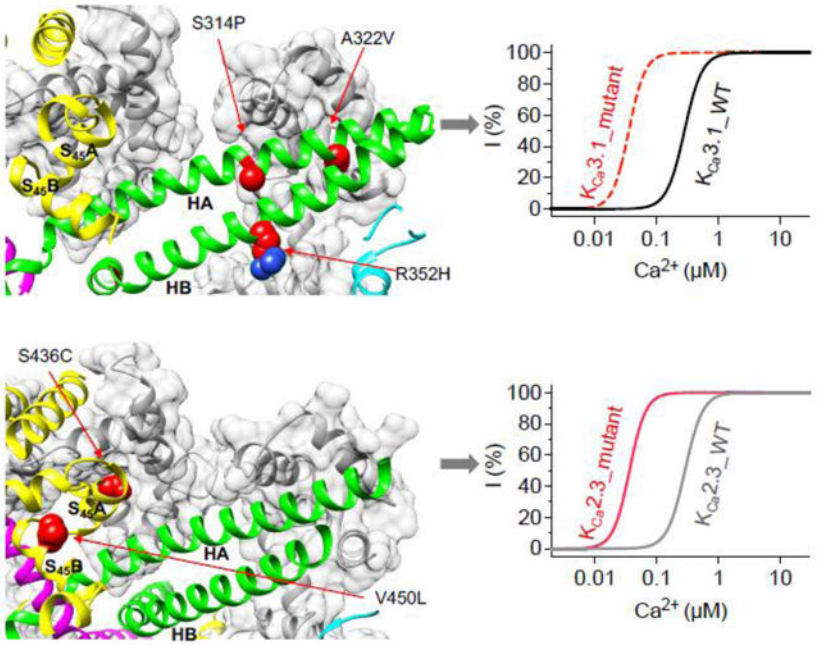
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$K_{Ca}3.1$_S314P, $K_{Ca}3.1$_A322V and $K_{Ca}3.1$_R352H channels with mutations in the HA/HB helices are hypersensitive to Ca^{2+}, whereas $K_{Ca}2.3$ channels with the equivalent mutations are not. The different effects of the equivalent mutations in the HA/HB helices on the apparent Ca^{2+} sensitivity of $K_{Ca}2.3$ and $K_{Ca}3.1$ channels may imply distinct modulation of the two channel subtypes by the HA/HB helices. AP14145 reduced the apparent Ca^{2+} sensitivity of the hypersensitive mutant $K_{Ca}2.3$ channels, suggesting the potential therapeutic usefulness of negative gating modulators.

Graphical Abstract



Keywords

HA/HB helices; $S_{45}A/S_{45}B$ helices; $K_{Ca}2.3$ and $K_{Ca}3.1$ channels; negative gating modulator

Introduction

Small- and intermediate-conductance Ca^{2+}-activated potassium ($K_{Ca}2.x$ and $K_{Ca}3.1$) channels are voltage independent. Mammalian *KCNN* genes encode three $K_{Ca}2.x$ channel subtypes, including $K_{Ca}2.1$ (SK1), $K_{Ca}2.2$ (SK2), $K_{Ca}2.3$ (SK3), and one $K_{Ca}3.1$ (IK or SK4) channel subtype. These channels are constitutively associated with the Ca^{2+}-binding protein calmodulin (CaM). Both $K_{Ca}2.x$ and $K_{Ca}3.1$ channels are activated by a unique Ca^{2+}-CaM gating mechanism.[1]

Human heterozygous genetic mutations of $K_{Ca}2.3$ channels have been linked to Zimmermann-Laband syndrome (ZLS)[2] and idiopathic noncirrhotic portal hypertension (INCPH)[3]. The ZLS-related (K269E, G350D and S436C) and INCPH-related (V450L) mutant channels exhibited faster kinetics of current activation by Ca^{2+} upon break-in with

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whole-cell patch-clamp recordings, and thus were inferred to have elevated apparent Ca^{2+} sensitivity[2]. The apparent Ca^{2+} sensitivity has not been determined quantitatively on any of these ZLS-related or INCPH-related mutant $K_{Ca}2.3$ channels. Meanwhile, heterozygous genetic mutations in $K_{Ca}3.1$, which underlies the so-called Gardos channel in erythrocytes, lead to hereditary xerocytosis. The Gardos channelopathy-related (V282M/E[4], S314P[5], A322V[6] and R352H[7–9]) mutations were also speculated to cause hypersensitivity to Ca^{2+} of $K_{Ca}3.1$ channels, mostly based on more prominent responses of mutant channel current to an inhibitor TRAM-34[5,7,8] or a positive modulator NS309[5,6]. The apparent Ca^{2+} sensitivity has only been determined quantitatively on one of these Gardos channelopathy-related mutant channels, $K_{Ca}3.1_R352H$[8].

A recent cryo-EM study revealed that the pore forming $K_{Ca}3.1$ channel subunits form a tetrameric structure, with the Ca^{2+} sensor protein CaM constitutively bound (Fig. 1A).[10] In each channel subunit, there are six transmembrane α-helical domains that are denoted S1–S6. CaM interacts with the HA/HB helices, in addition to the linker between the S4 and S5 transmembrane domains (S4–S5 linker).[10] The S4–S5 linker consists of two α-helices, $S_{45}A$ and $S_{45}B$ (Fig. 1B). The HA/HB helices from one channel subunit (green), the $S_{45}A$ and $S_{45}B$ helices from a neighboring channel subunit (yellow) and CaM (grey) closely interact with each other. Three of the Gardos channelopathy-related (S314P[5], A322V[6] and R352H[7–9]) missense mutations are located in the HA/HB helices, while the fourth one (V282M/E[4]) is located in the transmembrane S5 domain close to the pore.

Utilizing this cryo-EM structure of human $K_{Ca}3.1$ (PDB code: 6CNN) as a template, we generated a homology model of the human $K_{Ca}2.3$ channel (Fig. 1C). The INCPH-related (V450L) and one of the ZLS-related (S436C) mutations are located in the $S_{45}A/S_{45}B$ helices (Fig. 1D), while the other two ZLS-related (K269E and G350D) mutations are found in the N-terminus and transmembrane domains, respectively.

We previously identified a valine-to-phenylalanine (V-to-F) mutation in the HA helix that causes Ca^{2+} hypersensitivity in the $K_{Ca}2.1$, $K_{Ca}2.2$ and $K_{Ca}2.3$ subtypes[11], but the equivalent V-to-F mutation did not alter the apparent Ca^{2+} sensitivity in $K_{Ca}3.1$ channels.[12] In contrast, an asparagine-to-alanine (N-to-A) mutation in the $S_{45}A/S_{45}B$ helices increased the apparent Ca^{2+} sensitivity in both $K_{Ca}2.2a$ and $K_{Ca}3.1$ channel subtypes.[12] Thus, we speculate that the differential modulation of $K_{Ca}2.3$ and $K_{Ca}3.1$ channels may result more from effects mediated by the HA/HB helices than the $S_{45}A/S_{45}B$ helices. Although $K_{Ca}2.x$ and $K_{Ca}3.1$ channels are activated by the same Ca^{2+}-CaM gating mechanism, they exhibit differences in their function, modulation and pharmacology.[1] The $K_{Ca}3.1$ structures[10] and $K_{Ca}2.x$ homology models[12–15] offer an opportunity to perform structure-function studies and examine the differences between the channel subtypes, which might reveal structural information useful for the design of subtype-selective biophysical tools[12] and pharmacological tools[16]. To the best of our knowledge, Ca^{2+}-hypersensitive human genetic mutations have only been reported in $K_{Ca}2.3$ and $K_{Ca}3.1$ channels, but not in $K_{Ca}2.1$ and $K_{Ca}2.2$ channels. Utilizing these disease-causing mutations in the HA/HB helices and the $S_{45}A/S_{45}B$ helices as biophysical tools, we examined whether mutations in corresponding positions might influence $K_{Ca}2.3$ and $K_{Ca}3.1$ channels differentially.

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Here, we quantified the elevated apparent Ca^{2+} sensitivity of $K_{Ca}2.3$ channels with the INCPH-related (V450L) and the ZLS-related (S436C) mutations in the $S_{45}A/S_{45}B$ helices. Their equivalent mutations in the $S_{45}A/S_{45}B$ helices of $K_{Ca}3.1$ channels also increased the apparent Ca^{2+} sensitivity. On the other hand, the Gardos channelopathy-related (S314P, A322V and R352H) mutations in the HA/HB helices elevated the apparent Ca^{2+} sensitivity of $K_{Ca}3.1$ channels, whereas their equivalent mutations in $K_{Ca}2.3$ channels did not change the apparent Ca^{2+} sensitivity. A negative gating modulator, AP14145, reduced the apparent Ca^{2+} sensitivity of the INCPH- and ZLS-related mutant $K_{Ca}2.3$ channels, suggesting its potential therapeutic usefulness.

Results

Mutations in the $S_{45}A/S_{45}B$ helices of $K_{Ca}2.3$ and $K_{Ca}3.1$ channels

We previously identified an asparagine-to-alanine (N-to-A) mutation in the $S_{45}A/S_{45}B$ helices that increased the apparent Ca^{2+} sensitivity in both $K_{Ca}2.2a$ and $K_{Ca}3.1$ channel subtypes.[12] We first investigated the equivalent N441A mutation, together with the ZLS-related S436C and INCPH-related V450L mutations in the $S_{45}A/S_{45}B$ helices of $K_{Ca}2.3$ channels (Fig. 2A). In the study first describing these disease-associated mutations[2], the kinetics of current activation by Ca^{2+} upon break-in during whole-cell patch-clamp recordings was used to infer an increase in apparent Ca^{2+} sensitivity. In order to quantify this increase in apparent Ca^{2+} sensitivity, we used the inside-out patch-clamp to measure the Ca^{2+}-dependent activation of the mutant channels heterologously expressed in HEK293 cells (Fig. S1). With the inside-out patch configuration, we were able to measure the responses of the channels to various Ca^{2+} concentrations from the same patch and determine the apparent Ca^{2+} sensitivity of the mutant channels. Compared with wild-type (WT) $K_{Ca}2.3$ channel (EC_{50} = 0.30 ± 0.025 μM, n = 8), the S436C, N441A and V450L mutations effectively elevated the $K_{Ca}2.3$ channel apparent Ca^{2+} sensitivity (Fig. 2B). EC_{50} values for Ca^{2+} were 0.087 ± 0.0058 μM (n = 6, $P < 0.0001$), 0.064 ± 0.0098 μM (n = 6, $P < 0.0001$), and 0.15 ± 0.013 μM (n = 6, $P < 0.0001$), respectively (Fig. 2C and Table 1).

We then tested their equivalent mutations in $K_{Ca}3.1$ channels (Fig. S2). Three equivalent mutations S181C, N186A and V195L also increased apparent Ca^{2+} sensitivity of $K_{Ca}3.1$ channels (Fig. 3A). EC_{50} values for Ca^{2+} of the three mutants were 0.10 ± 0.011 μM (n = 7, $P < 0.0001$), 0.10 ± 0.0055 μM (n = 5, $P < 0.0001$), and 0.20 ± 0.020 μM (n = 5, $P < 0.01$), compared with WT $K_{Ca}3.1$ channels (EC_{50} = 0.27 ± 0.012 μM, n = 8, Fig. 3B and Table 1).

Mutations in the HA/HB helices of $K_{Ca}2.3$ and $K_{Ca}3.1$ channels

In our previous study, a valine-to-phenylalanine (V-to-F) mutation in the HA helix caused Ca^{2+} hypersensitivity in the $K_{Ca}2.1$, $K_{Ca}2.2$ and $K_{Ca}2.3$ subtypes[11], but the equivalent V-to-F mutation did not alter the apparent Ca^{2+} sensitivity in $K_{Ca}3.1$ channels.[12] Interestingly, a heterozygous V-to-F mutation in $K_{Ca}2.3$ (V555F) in a young child linked to ZLS was subsequently reported.[17] We here investigated this newly reported ZLS-related mutation hypothesizing that it would display increased apparent Ca^{2+} sensitivity.

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We investigated the $K_{Ca}3.1$_V298F mutation, together with the Gardos channelopathy-related (S314P[5], A322V[6] and R352H[7–9]) missense mutations located in the HA/HB helices of $K_{Ca}3.1$ channels (Fig. 4A and Fig. S3). Three Gardos channelopathy-related S314P, A322V and R352H mutations increased the apparent Ca^{2+} sensitivity of $K_{Ca}3.1$ channels (Fig. 4B). The EC_{50} values for Ca^{2+} of the three mutants were 0.064 ± 0.0033 μM (n = 9, P < 0.0001), 0.059 ± 0.0039 μM (n = 8, P < 0.0001), and 0.085 ± 0.0081 μM (n = 8, P < 0.0001), compared with WT $K_{Ca}3.1$ channels (EC_{50} = 0.27 ± 0.012 μM, n = 8, Fig. 4C and Table 1). Consistent with our previous report[12], the V298F mutation did not change the apparent Ca^{2+} sensitivity compared with the WT $K_{Ca}3.1$ channels (Fig. 4C and Table 1).

We next tested the equivalent mutations in $K_{Ca}2.3$ channels (Fig. S4). Three equivalent mutations A571P, T579V and R612H did not alter the $K_{Ca}3.1$ channel apparent Ca^{2+} sensitivity (Fig. 5A). The EC_{50} values for Ca^{2+} of the three mutants were 0.30 ± 0.027 μM (n = 8, P > 0.9999), 0.25 ± 0.032 μM (n = 6, P = 0.5666), and 0.27 ± 0.014 μM (n = 11, P = 0.8339), compared with WT $K_{Ca}2.3$ channels (EC_{50} = 0.30 ± 0.025 μM, n = 8, Fig. 5B and Table 1). On the other hand, the V555F mutation increased the apparent Ca^{2+} sensitivity compared with the WT $K_{Ca}2.3$ channels (Fig. 5B and Table 1).

We then tested another two ZLS-related mutations, K269E in the N-terminus (Fig. 6A) and G350D in the transmembrane domains (Fig. 6B) of $K_{Ca}2.3$ channels. Both K269E and G350D mutations increased the apparent Ca^{2+} sensitivity of $K_{Ca}2.3$ channels, with EC_{50} values of 0.086 ± 0.013 μM (n = 6, P < 0.0001) and 0.12 ± 0. 013 μM (n = 8, P < 0.0001), respectively (Table 1).

Effects of a negative gating modulator on the mutant $K_{Ca}2.3$ channels

AP30663[18], a negative gating modulator of $K_{Ca}2.3$ channels, is currently in early clinical development for conversion of atrial fibrillation[19]. AP30663 is not commercially available. To explore potential pharmacological therapies for the diseases caused by mutant $K_{Ca}2.3$ channels, we tested the effects of a structurally related compound, AP14145[20], on the WT and mutant $K_{Ca}2.3$ channels (Fig. 6A–F). AP14145 (10 μM) right-shifted the Ca^{2+} concentration-response curves of all the mutant and WT channels, indicating its effectiveness as a negative gating modulator. AP14145 induced a ~3.6-fold decrease in the apparent Ca^{2+} sensitivity of $K_{Ca}2.3$_WT channels (Fig. 6F, Table 1). Its effects on the mutant channels were somewhat less prominent. AP14145 changed the EC_{50} values for Ca^{2+} of $K_{Ca}2.3$_K269E, $K_{Ca}2.3$_G350D, $K_{Ca}2.3$_S436C, $K_{Ca}2.3$_V450L and $K_{Ca}2.3$_V555F channels by ~1.63, ~2.25, ~2.18, ~2.07 and ~2.09 folds, respectively (Fig. 6A–E, Table 1).

Discussion

$K_{Ca}2.x/K_{Ca}3.1$ channels are a unique group of potassium channels that are voltage-independent.[21] Mutations in both $K_{Ca}2.3$ and $K_{Ca}3.1$ channels have been linked with genetic human disorders. We here quantified the effects of these mutations on the apparent Ca^{2+} sensitivity of the channels. The Gardos channelopathy-related $K_{Ca}3.1$_S314P[5], $K_{Ca}3.1$_A322V[6] and $K_{Ca}3.1$_R352H[7–9] mutations in the HA/HB helices

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caused hypersensitivity to Ca^{2+}, which could lead to increased K^+ outflow and erythrocyte dehydration[5,7,22,23].

The ZLS- and INCPH-related mutations in the $S_{45}A/S_{45}B$ helices increased the apparent Ca^{2+} sensitivity of $K_{Ca}2.3$ channels. Exactly how the Ca^{2+} hypersensitivity of $K_{Ca}2.3$ channels leads to ZLS and INCPH is unclear. Increased $K_{Ca}2.3$ channel function in the central nervous system might be related to the defective cognitive function.[24] Elevated $K_{Ca}2.3$ channel activity in the vascular endothelium is speculated to be involved in distal digital hypoplasia[2]. A previous report[2] demonstrated faster time courses of channel activation upon breakthrough of the cell membrane to the whole-cell configuration with pipette solution containing 1 μM of Ca^{2+}, implying higher apparent Ca^{2+} sensitivity of the ZLS- and INCPH-related mutant $K_{Ca}2.3$ channels. In this study, we quantitatively determined the EC_{50} values for Ca^{2+} of these mutant channels, using the inside-out configuration (Table 1). They are indeed hypersensitive to Ca^{2+}, with the ZLS-related mutants being more sensitive to Ca^{2+} than the INCPH-related mutant channel. The reported ZLS[2] and INCPH[3] patients have heterozygous mutations in their *KCNN3* alleles. Their $K_{Ca}2.3$ channels might be comprised of both WT and mutant subunits in the tetrameric channel architecture, which may be speculated to have apparent Ca^{2+} sensitivity higher than the WT but lower than the mutant channels reported in this study (Table 1).

The amino acid sequences of $K_{Ca}2.3$ at the $S_{45}A$ and $S_{45}B$ helices shares ~59% identity with the $K_{Ca}3.1$ channel (highlighted in yellow, Fig. 2A). These two channel subtypes share ~60% identity at the HA/HB helices (highlighted in green, Fig. 4A). When bound with Ca^{2+}, CaM forms a stable structural group with the HA/HB helices, and the $S_{45}A/S_{45}B$ helices from a neighboring $K_{Ca}3.1$ channel subunit (Fig. 1A,B)[10]. A homology model of the human $K_{Ca}2.3$ channels showed similar interactions (Fig. 1C,D). Our previous study addressed the crucial role of the interactions between CaM, the HA/HB helices and the $S_{45}A/S_{45}B$ helices for the apparent Ca^{2+} sensitivity of the channels.[12]

In this study, we demonstrated that equivalent mutations in the $S_{45}A/S_{45}B$ helices of $K_{Ca}2.3$ ($K_{Ca}2.3$_S436C, $K_{Ca}2.3$_N441A and $K_{Ca}2.3$_V450L) and $K_{Ca}3.1$ ($K_{Ca}3.1$_S181C, $K_{Ca}3.1$_N186A and $K_{Ca}3.1$_V195L) channels elevated the apparent Ca^{2+} sensitivity of both channel subtypes (Fig. 2 and Fig. 3). In contrast, mutations in the HA/HB helices of $K_{Ca}3.1$ ($K_{Ca}3.1$_S314P, $K_{Ca}3.1$_A322V and $K_{Ca}2.3$_R352H) channels caused Ca^{2+} hypersensitivity (Fig. 4), whereas their equivalent mutations in $K_{Ca}2.3$ ($K_{Ca}2.3$_A571P, $K_{Ca}2.3$_T579V and $K_{Ca}2.3$_R612H) did not alter the apparent Ca^{2+} sensitivity (Fig. 5). Furthermore, a mutation in the HA/HB helices of $K_{Ca}2.3$ ($K_{Ca}2.3$_V555F) channels caused Ca^{2+} hypersensitivity (Fig. 5), whereas its equivalent mutations in $K_{Ca}3.1$ ($K_{Ca}3.1$_V298F) did not change the apparent Ca^{2+} sensitivity (Fig. 4). As such, more synchronized changes in the apparent Ca^{2+} sensitivity of $K_{Ca}2.3$ and $K_{Ca}3.1$ channel subtypes were observed in response to equivalent mutations in the $S_{45}A/S_{45}B$ helices. In contrast, these two channel subtypes often exhibited differential responses to equivalent mutations in the HA/HB helices. Given the involvement of $K_{Ca}2.3$ and $K_{Ca}3.1$ channels in genetic disorders, understanding the regulation of their apparent Ca^{2+} sensitivity is critical. With cryo-EM structures only available for $K_{Ca}3.1$ channels[10], caution will be needed when performing structure-function studies of $K_{Ca}2.x$ channel subtypes, especially for the HA/HB helices.

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It was reported that the apparent Ca^{2+} sensitivity of $K_{Ca}2.2$ channels can be negatively regulated by the phosphorylation of CaM.[25] Indeed, the apparent Ca^{2+} sensitivity of all $K_{Ca}2.x/K_{Ca}3.1$ channel subtypes can be reduced by the phosphorylation of CaM.[26] As such, the EC_{50} values for Ca^{2+} determined in HEK293 cells may deviate from the apparent Ca^{2+} sensitivity of the channels expressed in other cell types (e.g. endothelial cells), as a result of the phosphorylation status.[26] $K_{Ca}3.1$ is unique among these four subtypes in that its activation can be enhanced by the phosphorylation of a histidine residue (H358) in the HB helix[27], possibly by antagonizing copper-mediated inhibition of the channel.[28] This unique regulation of $K_{Ca}3.1$ by the phosphorylation of H358 in the HB helix may also echo this notion that $K_{Ca}3.1$ channels are modulated differently by the HA/HB helices from $K_{Ca}2.x$ channels.

The pharmacology for $K_{Ca}3.1$ channel subtype has been well developed[1]. Senicapoc[29] and TRAM-34[30] inhibit $K_{Ca}3.1$ channels with IC_{50} values of ~11 nM and ~20 nM, respectively, with selectivity (200–1000-fold) towards IK channels over other channel subtypes. With excellent pharmacokinetic properties in humans, senicapoc has become a potential candidate for the treatment of Gardos channelopathy, a subset of hereditary xerocytosis.[31] A clinical trial (ClinicalTrials.gov Identifier: NCT04372498) is studying senicapoc in patients with hereditary xerocytosis caused by $K_{Ca}3.1$ channel mutations. On the other hand, no pharmacological treatment has been explored for ZLS and INCPH caused by $K_{Ca}2.3$ channel mutations. AP14145, which is equipotent in modulating $K_{Ca}2.2$ and $K_{Ca}2.3$[20], with no effect on $K_{Ca}3.1$, has been reported to treat atrial fibrillation in pigs[32] and goats[33]. A structurally related compound, AP30663[18], is currently in early clinical development for conversion of atrial fibrillation[19]. Our results indicated that AP14145 right-shifted the Ca^{2+} concentration-response curves (Fig. 6) and reduced the apparent Ca^{2+} sensitivity (Table 1) of the mutant $K_{Ca}2.3$ channels. The $K_{Ca}2.3_V450L$ mutant channel with an EC_{50} value for Ca^{2+} of 0.15 ± 0.013 μM (Table 1) is associated with INCPH. This patient with $K_{Ca}2.3_V450L$ mutation was not classified as ZLS[3]. Among the ZLS-related mutant channels, $K_{Ca}2.3_G350D$ is the least sensitive to Ca^{2+} with an EC_{50} value of 0.12 ± 0.013 μM (Table 1). As such, it can be inferred that mutations that elevate apparent Ca^{2+} sensitivity to < 0.12 μM can be associated with ZLS. AP14145 was able to reduce the apparent Ca^{2+} sensitivity of the ZLS-related mutant channels to 0.14 ± 0.011 μM ($K_{Ca}2.3_K269E$ + AP14145), 0.27 ± 0.025 μM ($K_{Ca}2.3_G350D$ + AP14145), 0.19 ± 0.026 μM ($K_{Ca}2.3_S436C$ + AP14145) and 0.14 ± 0.015 μM ($K_{Ca}2.3_V555F$ + AP14145, Table 1), implying the potential therapeutic usefulness of negative $K_{Ca}2$ channel gating modulators. The *KCNN3* gene encoding $K_{Ca}2.3$ channels has only very recently been associated with ZLS[2] and INCPH[3]. To the best of our knowledge, there are currently no transgenic mouse models for these disorders. Future development of animal models will enable preclinical studies for therapeutic options.

Materials and Methods

Electrophysiology

The effect of mutations on the apparent Ca^{2+} sensitivity of $K_{Ca}2.3$ and $K_{Ca}3.1$ channels was investigated as previously described.[12,15,26] Briefly, mutations were introduced to

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human $K_{Ca}2.3$ and human $K_{Ca}3.1$ channels using QuickChange II site-directed mutagenesis kit (Agilent) or through molecular cloning services (Genscript). The WT and mutant channel cDNAs, constructed in the pIRES2-AcGFP1 vector (Clontech) were transfected into HEK293 cells by the calcium–phosphate method. AP14145 was purchased from Tocris Bioscience. $K_{Ca}2.3$ and $K_{Ca}3.1$ currents were recorded 1–2 days after transfection, with an Axon200B amplifier (Molecular Devices) at room temperature.

pClamp 10.5 (Molecular Devices) was used for data acquisition and analysis. The resistance of the patch electrodes ranged from 3–5 MΩ. The pipette solution contained (in mM): 140 KCl, 10 Hepes (pH 7.4), 1 $MgSO_4$. The bath solution containing (in mM): 140 KCl, 10 Hepes (pH 7.2), 1 EGTA, 0.1 Dibromo-BAPTA, and 1 HEDTA was mixed with Ca^{2+} to obtain the desired free Ca^{2+} concentrations, calculated using the software by Chris Patton of Stanford University (https://somapp.ucdmc.ucdavis.edu/pharmacology/bers/maxchelator/webmaxc/webmaxcS.htm). The Ca^{2+} concentrations were verified using a Ca^{2+} calibration buffer kit (Thermo Fisher Scientific). Briefly, a standard curve was generated using the Ca^{2+} buffers from the kit and a fluorescence Ca^{2+} indicator. Then the Ca^{2+} concentrations of our bath solutions were determined through interpolation on the standard curve.

Currents were recorded using an inside-out patch configuration. The inside-out patch configuration allows us to measure the response of the channels to various Ca^{2+} concentrations from the same patch and effectively compare the apparent Ca^{2+} sensitivity of WT and mutant channels. The intracellular face was initially exposed to a zero-Ca^{2+} bath solution, and subsequently to bath solutions with a series of Ca^{2+} concentrations. Currents were recorded by repetitive 1-s-voltage ramps from − 100 mV to + 100 mV from a holding potential of 0 mV. One minute after switching of bath solutions, ten sweeps with a 1-s interval were recorded. The integrity of the patch was examined by switching the bath solution back to the zero-Ca^{2+} buffer. Data from patches, which did not show significant changes in the seal resistance after solution changes, were used for further analysis. To construct the concentration-dependent positive modulation of channel activities, the current amplitudes at − 90 mV in response to various concentrations of Ca^{2+} were normalized to that obtained at maximal concentration of Ca^{2+}. The normalized currents were plotted as a function of the concentrations of Ca^{2+}. EC_{50} values and Hill coefficients were determined by fitting the data points to a standard concentration–response curve ($Y = 100/(1 + (X/EC_{50})^{\wedge} -$ Hill)). The data analysis was performed using pClamp 10.5 (Molecular Devices) in a blinded fashion. All data are shown as mean ± SEM unless otherwise indicated. One-way ANOVA and Tukey's post hoc tests were used for data comparison of three or more groups. The Student's *t*-test was used for data comparison if there were only two groups.

Computational modeling

We built homology model for the human $K_{Ca}2.3$ channel based on a cryo-EM structure of $K_{Ca}3.1$ channel (PDB Code: 6CNN). Sequence alignment among $K_{Ca}2.3$ and $K_{Ca}3.1$ channels was generated by Clustal Omega server (https://www.ebi.ac.uk/Tools/msa/clustalo/). The sequence identity between $K_{Ca}2.3$ and $K_{Ca}3.1$ is 46.6%, which makes the $K_{Ca}3.1$ Cryo-EM structure an excellent structural template to generate homology models for the $K_{Ca}2.3$ channels. We used the MODELLER program[34] to generate 10 initial homology

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models for $K_{Ca}2.3$ channel based on the $K_{Ca}3.1$ structural template, and selected the one with the best internal DOPE score from the program.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

We are grateful to Misa Nguyen, Young Hur, and Lucia Basilio for technical assistance. M.Z. was supported by a Scientist Development Grant 13SDG16150007 from American Heart Association, a YI-SCA grant from National Ataxia Foundation and a grant 4R33NS101182-03 from NIH.

Abbreviations

N-to-A	Asparagine-to-alanine
Ca^{2+}	Calcium
CaM	Calmodulin
INCPH	Idiopathic noncirrhotic portal hypertension
$K_{Ca}3.1$ or IK	Intermediate-conductance Ca^{2+}-activated K^+ channels
$K_{Ca}2.x$ or SK	Small-conductance Ca^{2+}-activated K^+ channels
V-to-F	Valine-to-phenylalanine
WT	Wild-type
ZLS	Zimmermann-Laband syndrome

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Highlights

- Disease-causing mutations elevated the apparent Ca^{2+} sensitivity of $K_{Ca}2.3$ channels
- Disease-causing mutations elevated the apparent Ca^{2+} sensitivity of $K_{Ca}3.1$ channels
- $K_{Ca}2.3$ and $K_{Ca}3.1$ channels are regulated differentially by the HA/HB helices
- AP14145 reduced the Ca^{2+} sensitivity of disease related mutant $K_{Ca}2.3$ channels

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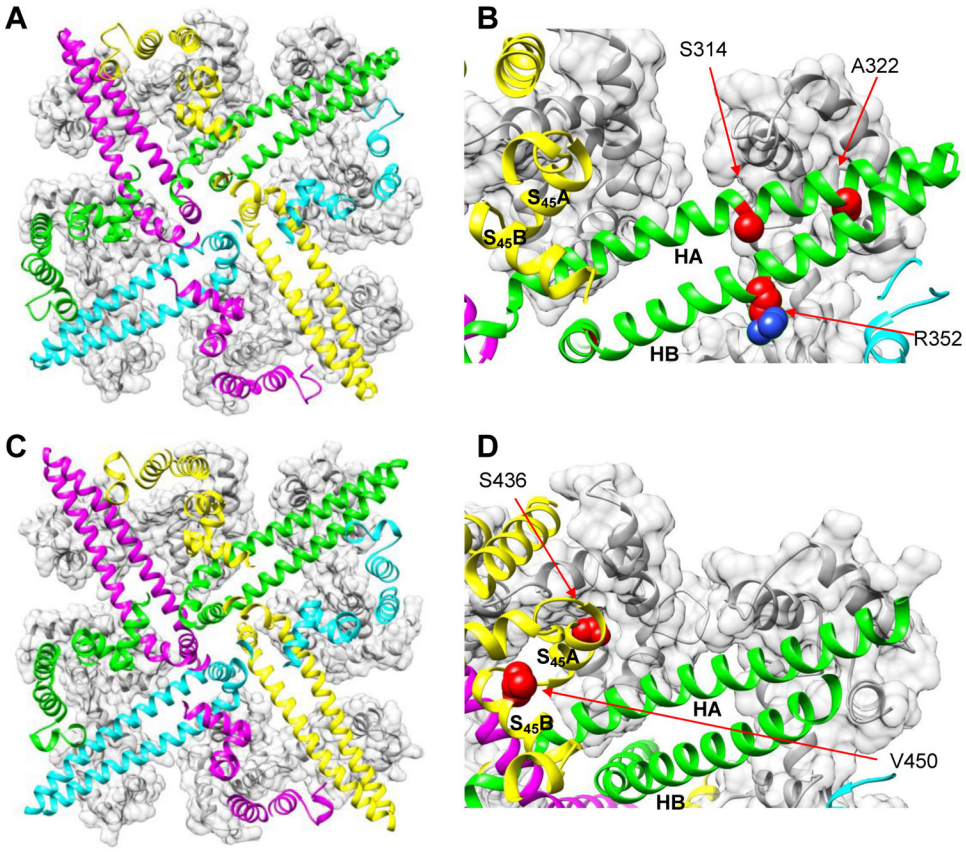


Fig. 1. Channelopathy-causing mutations in the $S_{45}A/S_{45}B$ helices and HA/HB helices of human $K_{Ca}2.3$ and human $K_{Ca}3.1$ channels.

(A) The $S_{45}A/S_{45}B$ helices and HA/HB helices of human $K_{Ca}3.1$ channel cryo-EM structure with Ca^{2+} (PDB: 6cnn). Four CaM are shown in grey. Four channel subunits are shown in different colors (green, yellow, cyan, and magenta). (B) Ca^{2+} bound CaM (grey), HA and HB helices from one channel subunit (green), the $S_{45}A$ and $S_{45}B$ helices from a neighboring channel subunit (yellow) become a stable structural group. The Gardos channelopathy-related (S314P, A322V and R352H) mutations are shown as spheres (carbon atoms shown in red and nitrogen atoms shown in blue). (C) The $S_{45}A/S_{45}B$ helices and HA/HB helices of a homology model of the human $K_{Ca}2.3$ channel was generated using the human $K_{Ca}3.1$ channel cryo-EM structure with Ca^{2+}. Four CaM are shown in grey. Four channel subunits are shown in different colors (green, yellow, cyan, and magenta). (D) The ZLS-related (S436C) and INCPH-related (V450L) mutations are shown as spheres (carbon atoms shown in red).

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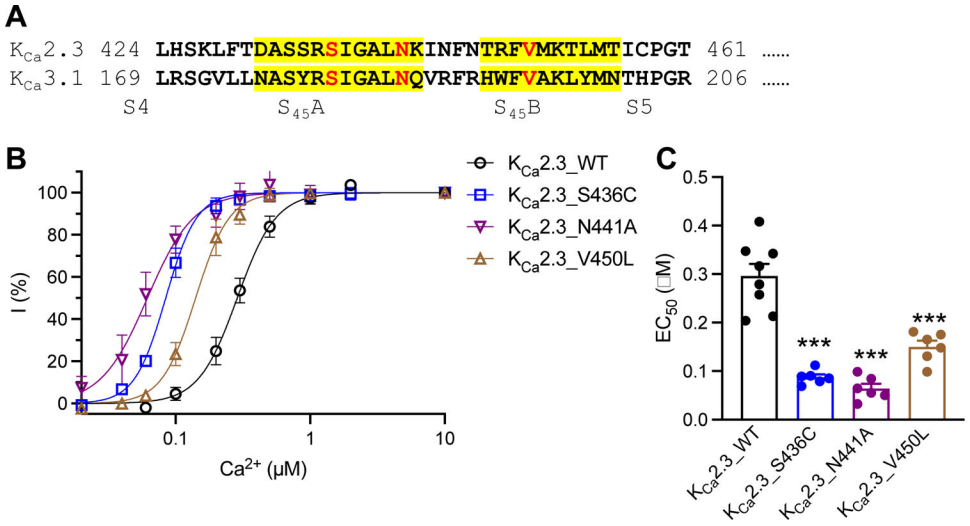


Fig. 2. The ZLS-related (S436C) and INCPH-related (V450L) mutations caused Ca^{2+} hypersensitivity of $K_{Ca}2.3$ channels.

(A) Amino acid sequence alignment of human $K_{Ca}2.3$ [GenBank: NP_002240.3] and human $K_{Ca}3.1$ [GenBank: NP_002241.1] at the $S_{45}A$ and $S_{45}B$ helices (highlighted in yellow). Amino acid residues mutated are shown in red font. (B) Activation by Ca^{2+} of the WT and mutant $K_{Ca}2.3$ channels carrying mutations in the $S_{45}A/S_{45}B$ helices. (C) EC_{50} values for activation by Ca^{2+} of the WT and mutant $K_{Ca}2.3$ channels. *** $P < 0.001$ compared with WT. No asterisk means no statistical significance compared with WT.

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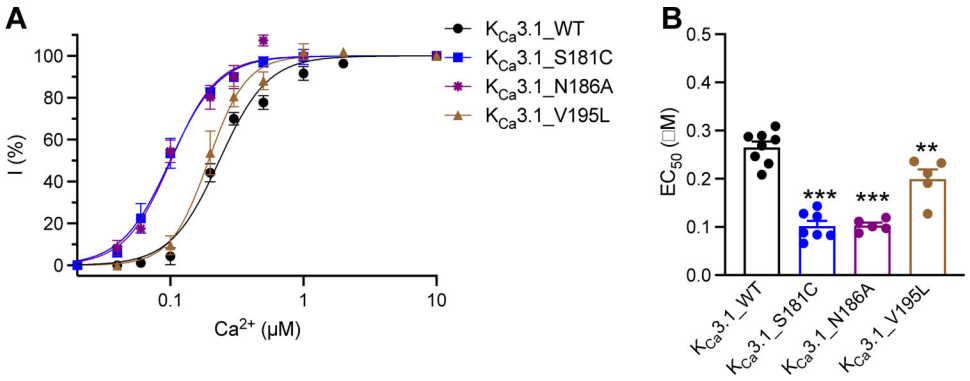


Fig. 3. The mutations equivalent to $K_{Ca}2.3$_S436C and $K_{Ca}2.3$_V450L caused Ca^{2+} hypersensitivity of $K_{Ca}3.1$ channels.

(A) Concentration-dependent activation by Ca^{2+} of the WT and mutant $K_{Ca}3.1$ channels carrying mutations in the $S_{45}A/S_{45}B$ helices. (B) EC_{50} values for the activation of the WT and mutant $K_{Ca}3.1$ channels by Ca^{2+}. ** $P < 0.01$, *** $P < 0.001$ compared with WT channels. No asterisk means no statistical significance compared with respective WT channels.

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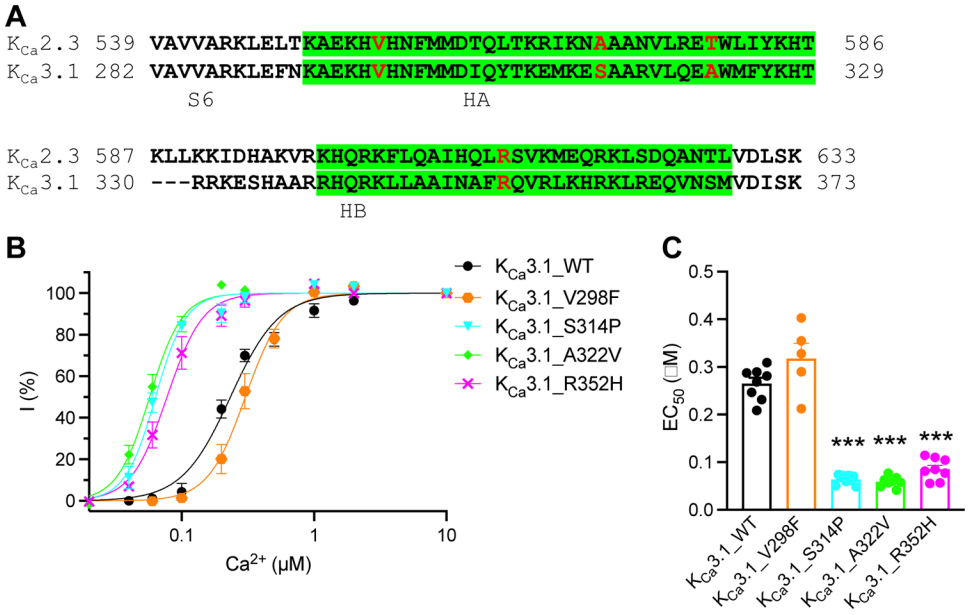


Fig. 4. The Gardos channelopathy-related (S314P, A322V and R352H) mutations elevated Ca^{2+} sensitivity of $K_{Ca}3.1$ channels.

(A) Amino acid sequence alignment of human $K_{Ca}2.3$ [GenBank: NP_002240.3] and human $K_{Ca}3.1$ [GenBank: NP_002241.1] at the HA and HB helices (highlighted in green). Amino acid residues mutated are shown in red font. (B) Concentration-dependent activation by Ca^{2+} of the WT and mutant $K_{Ca}3.1$ channels carrying mutations in the HA/HB helices. (C) EC_{50} values for the activation of the WT and mutant $K_{Ca}3.1$ channels by Ca^{2+}. *** $P < 0.001$ compared with WT channels. No asterisk means no statistical significance compared with respective WT channels.

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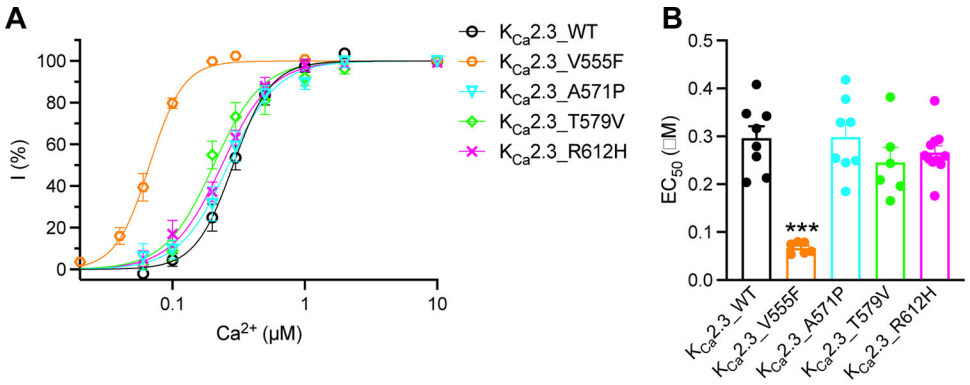


Fig. 5. The mutations equivalent to K_{Ca}3.1_S314P, K_{Ca}3.1_A322V and K_{Ca}3.1_R352H did not change apparent Ca^{2+} sensitivity of K_{Ca}2.3 channels.

(A) Activation by Ca^{2+} of the WT and mutant K_{Ca}2.3 channels carrying mutations in the HA/HB helices. (B) EC_{50} values for activation by Ca^{2+} of the WT and mutant K_{Ca}2.3 channels. *** $P < 0.001$ compared with WT. No asterisk means no statistical significance compared with WT.

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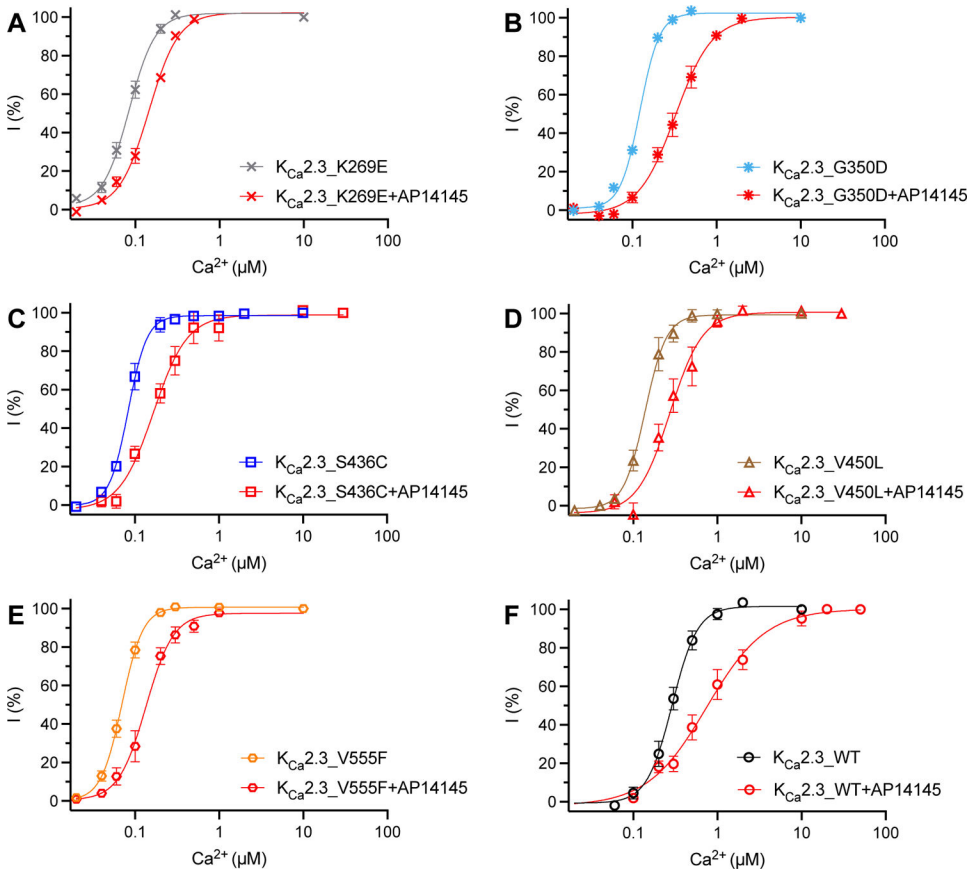


Fig. 6. The effects of AP14145 on the Ca^{2+}-dependent activation of WT and mutant $K_{Ca}2.3$ channels.

Activation by Ca^{2+} of the $K_{Ca}2.3$_K269E (A), $K_{Ca}2.3$_G350D (B), $K_{Ca}2.3$_S436C (C), $K_{Ca}2.3$_V450L (D), $K_{Ca}2.3$_V555F (E) and $K_{Ca}2.3$_WT (F) channels.

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Table 1.

Apparent Ca^{2+} sensitivity of WT and mutant channels.

$K_{Ca}2.3$	EC_{50} for Ca^{2+} (μM)	Hill coefficient	Statistics	$K_{Ca}3.1$	EC_{50} for Ca^{2+} (μM)	Hill coefficient	Statistics
$K_{Ca}2.3$_WT	0.30 ± 0.025	3.88 ± 0.59	NA	$K_{Ca}3.1$_WT	0.27 ± 0.012	2.87 ± 0.48	N/A
$K_{Ca}2.3$_WT + AP14145	1.08 ± 0.10	1.34 ± 0.14	P < 0.0001 *vs* $K_{Ca}2.3$_WT				
$K_{Ca}2.3$_K269E	0.086 ± 0.013	3.45 ± 0.46	P < 0.0001 *vs* $K_{Ca}2.3$_WT				
$K_{Ca}2.3$_K269E + AP14145	0.14 ± 0.011	3.15 ± 0.72	P < 0.05 *vs* $K_{Ca}2.3$_K269E				
$K_{Ca}2.3$_G350D	0.12 ± 0.013	4.19 ± 0.43	P < 0.0001 *vs* $K_{Ca}2.3$_WT				
$K_{Ca}2.3$_G350D + AP14145	0.27 ± 0.025	3.86 ± 1.09	P < 0.0001 *vs* $K_{Ca}2.3$_G350D				
$K_{Ca}2.3$_S436C	0.087 ± 0.0058	4.37 ± 0.85	P < 0.0001 *vs* $K_{Ca}2.3$_WT	$K_{Ca}3.1$_S181C	0.10 ± 0.011	2.78 ± 0.25	P < 0.0001 *vs* $K_{Ca}3.1$_WT
$K_{Ca}2.3$_S436C + AP14145	0.19 ± 0.026	2.28 ± 0.45	P < 0.01 *vs* $K_{Ca}2.3$_S436C				
$K_{Ca}2.3$_N441A	0.064 ± 0.0098	4.21 ± 1.09	P < 0.0001 *vs* $K_{Ca}2.3$_WT	$K_{Ca}3.1$_N186A	0.10 ± 0.0055	2.66 ± 0.45	P < 0.0001 *vs* $K_{Ca}3.1$_WT
$K_{Ca}2.3$_V450L	0.15 ± 0.013	4.15 ± 0.58	P < 0.001 *vs* $K_{Ca}2.3$_WT	$K_{Ca}3.1$_V195L	0.20 ± 0.020	4.08 ± 0.89	P < 0.01 *vs* $K_{Ca}3.1$_WT
$K_{Ca}2.3$_V450L + AP14145	0.31 ± 0.061	3.01 ± 0.62	P < 0.05 *vs* $K_{Ca}2.3$_V450L				
$K_{Ca}2.3$_V555F	0.067 ± 0.0047	3.77 ± 0.20	P < 0.0001 *vs* $K_{Ca}2.3$_WT	$K_{Ca}3.1$_V298F	0.32 ± 0.032	3.39 ± 0.43	P > 0.05 *vs* $K_{Ca}3.1$_WT
$K_{Ca}2.3$_V555F + AP14145	0.14 ± 0.015	2.81 ± 0.33	P < 0.001 *vs* $K_{Ca}2.3$_V555F				
$K_{Ca}2.3$_A571P	0.30 ± 0.027	2.75 ± 0.25	P > 0.9999 *vs* $K_{Ca}2.3$_WT	$K_{Ca}3.1$_S314P	0.064 ± 0.0033	4.07 ± 0.41	P < 0.0001 *vs* $K_{Ca}3.1$_WT
$K_{Ca}2.3$_T579V	0.25 ± 0.032	2.79 ± 0.47	P = 0.5666 *vs* $K_{Ca}2.3$_WT	$K_{Ca}3.1$_A322V	0.059 ± 0.0039	3.94 ± 0.29	P < 0.0001 *vs* $K_{Ca}3.1$_WT
$K_{Ca}2.3$_R612H	0.27 ± 0.014	2.78 ± 0.23	P = 0.8339 *vs* $K_{Ca}2.3$_WT	$K_{Ca}3.1$_R352H	0.085 ± 0.0081	3.66 ± 0.55	P < 0.0001 *vs* $K_{Ca}3.1$_WT